

EX.NO: 8  
-3-2026

## TRANSIENT ANALYSIS OF RLCCIRCUIT

AIM:

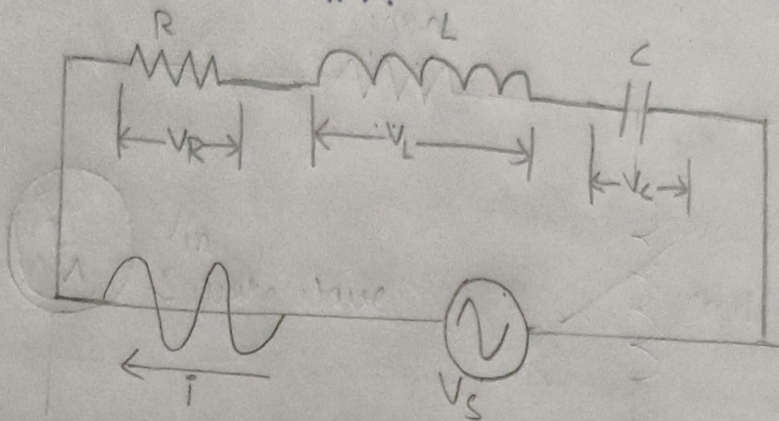
To determine transient response of a Series RLCCircuit using Multisim Simulator.

APPARATUS REQUIRED:

| S.NO | COMPONENTS                 | SPECIFICATION | QUANTITY |
|------|----------------------------|---------------|----------|
| 1)   | CRO                        | 10MHz         | 1        |
| 2)   | DIB                        |               |          |
| 3)   | Resistors                  | 1K- $\Omega$  |          |
| 4)   | Bread Board                | -             | 1        |
| 5)   | Connecting Wires           | -             | -        |
| 6)   | PC with Multisim Simulator | -             | -        |



# CIRCUIT DIAGRAM:



## TABULATION:

| Time(t)   | Observed Values          |                         |                         |
|-----------|--------------------------|-------------------------|-------------------------|
|           | Voltage across Capacitor | Voltage Across Inductor | Voltage Across Resistor |
| $t=0$     | 0V                       | 5V                      | 0V                      |
| $t=\tau$  | 2.5V                     | 3.0V                    | 2V                      |
| $t=2\tau$ | 3.6V                     | 1.8V                    | 3V                      |
| $t=3\tau$ | 4.3V                     | 1.0V                    | 4V                      |
| $t=4\tau$ | 4.7V                     | 0.5V                    | 4.5V                    |
| $t=5\tau$ | 5V                       | 0V                      | 5V                      |

plg verified



TABULATION: (Both Capacitor and Inductor)

$V_{in}$  Amplitude : 5V

$V_{in}$  Frequency : 1KHz

Time Constant  $\tau = \frac{R_C}{R} = 1ms$

Time Constant  $\tau = L/R = 1ms$

Time(t)  $V_L(t)$  - Voltage Across Inductor

Observed Value

Calculated Value

Given CALCULATION:

Given :  $R = 10k\Omega$ ,  $C = 0.1\mu F$ ,  $L = 10mH$

Time Constant

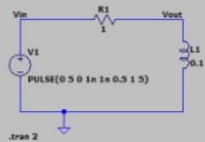
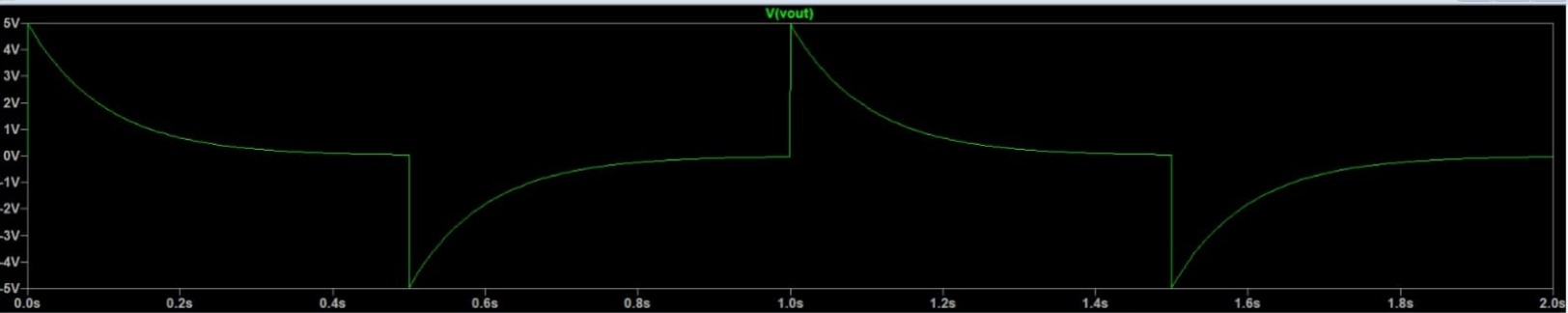
RC Time Constant

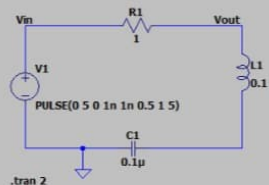
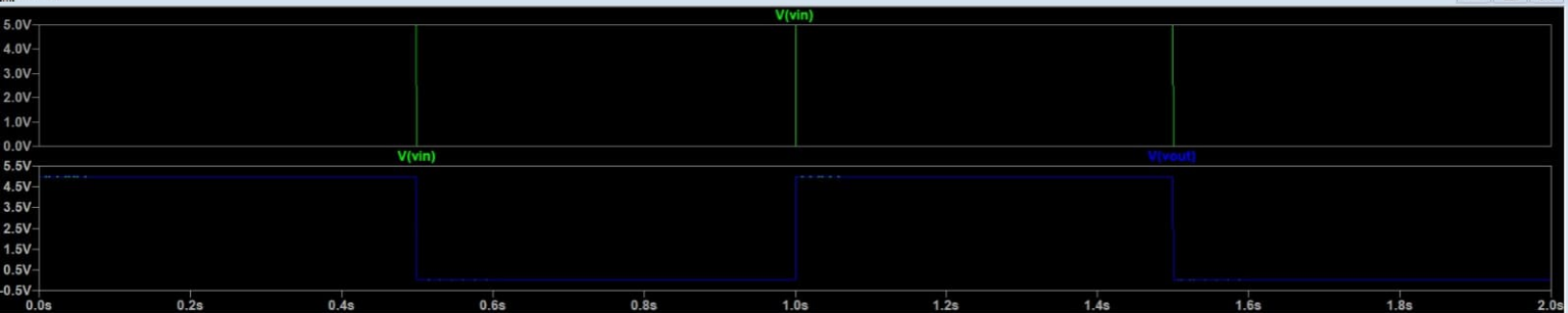
$$\tau = R \times C$$

$$= 10,000 \times 0.1 \times 10^{-6} = 0.001s = 1ms$$

RL Time Constant

$$\tau = \frac{L}{R} = \frac{0.01}{10000} = 1ms$$







### PROCEDURE :

1) Make Connections as per the Circuit Diagram

2) Calculate the Values of  $V_L(t)$  and write in Table.

3) Compare the Calculated Values and Measured Values of Voltages in Both the Cases

MARKS SPILT UP :-

| S.No | Description | Marks Alloted | Marks Obtained |
|------|-------------|---------------|----------------|
| 1)   | Experiment  | 20 Marks      | 20             |
| 2)   | Observation | 20 Marks      | 20             |
| 3)   | Calculation | 20 Marks      | 20             |
| 4)   | Viva        | 20 Marks      | 20             |
| 5)   | Record      | 20 Marks      | 20             |
|      |             | Total         | 100            |

10/12

### RESULT :

Thus, The ~~Transient~~ Analysis of RLC Circuit is done practically using Multisim Simulator.